A3 Project Proposal

Master in Health Data Science — MHEDAS

Subject: Complex Networks

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Group: T

Delivery: December 14, 2025

Option 1:

Title: The Architecture of Guardiola's Tiki-Taka: A Complex Network Analysis of the 2011 UCL Final

Description: The 2011 Champions League Final between FC Barcelona and Manchester United is widely regarded as a masterpiece of tactical football, showcasing the pinnacle of Guardiola's tiki-taka philosophy. This project aims to study the match as a Complex Network by modeling player interactions as a directed graph, where nodes represent players and edges represent passes. We aim to quantify the structural properties that defined Guardiola's dominance. We hypothesize that his system exhibited specific topological features: high clustering coefficients and robust small-world properties that distinguished it from the inefficient structure of Manchester United. Furthermore, we aim to analyze the specific role of Lionel Messi as a False Nine hub, whose positioning and passing patterns were crucial to unlocking defenses.

Data: To conduct this analysis, we will utilize event data from the 2011 Champions League Final, sourced from StatsBomb. This dataset provides event-level information, including pass locations, timestamps, and player identities.

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Option 2:

Title: Sophie's Complex World: A Network Analysis of Gaarder's Philosophical Novel

Description: *El Mundo de Sofía* (Sophie's World) is a novel written by Jostein Gaarder, who describes the history of philosophy through the narration of a teenage girl's life. We aim to describe the story from a complex network perspective, representing each character as a node, with edges between them defined if two characters are mentioned within a predefined word window. By analyzing the network, we would try to describe communities formed by different philosophers. We also want to identify important individuals, like Alberto Knox, who acts as the guide throughout the story.

Data: To conduct this analysis, we will utilize the full digital text of *El Mundo de Sofía* in plain text format. We will employ Natural Language Processing (NLP) techniques to extract entity interactions and build the adjacency matrix for the network analysis.

Option 3:

Title: Mapping the Multiverse: Community Detection in the Marvel Social Network

Description: The Marvel Universe consists of a vast interconnected Multiverse of narratives consisting of over 8,000 characters and decades of crossovers. The idea would be about analyzing this vast lore as a social collaboration network. We will construct a graph where nodes represent characters and edges are defined by co-occurrence in the same comic. Our goal is to map the topology of this Multiverse using Centrality Measures (Degree and Betweenness) to determine if icons like Captain America are truly the structural hubs (like in the movies), or if characters act as the primary bridges between different storylines (e.g., connecting the Avengers cluster to the X-Men cluster). We will use Community Detection algorithms to see if the structure of the graph naturally reproduces the teams without prior knowledge of the lore.

Data: We will utilize the open-source Marvel Universe Social Network dataset, which maps character appearances across decades of publications.